Real-Time eBPF/XDP-Based Deep Learning Framework for Anomaly Traffic Detection in Internet of Things Networks

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Internet of Things (IoT) networks face resource constraints yet require real-time anomaly detection across diverse protocols. We propose a dynamic, feature-selective framework for heterogeneous IoT networks. It integrates an extended Berkeley packet filter/express data path (eBPF/XDP) in the driver path with a pretrained lightweight deep learning model in user space, enabling two-stage adaptive feature selection. The feature-selective engine dynamically identifies the most discriminative traffic characteristics under evolving attacks. We validate generalizability using CICDDoS2019, IoT-23, and TON_IoT datasets. For example, on CICDDoS2019, our model trims 77 features down to only five while maintaining nearly identical accuracy (93.3% versus 92.73%). On a Raspberry Pi 4B, packet interception averages 1.85 µs and inference 10.38 ms, with negligible CPU overhead. Results confirm both responsiveness and accuracy, addressing security challenges in IoT networks and edge computing systems.

With the increasing sophistication of cyberattacks, Internet of Things (IoT) environments and low-resource edge computing nodes face significant challenges in anomaly traffic detection. Several security-related factors contribute to this complexity, for example:

- *Resource constraints:* IoT devices (e.g., sensors) operate with limited CPU power, memory, limited battery life, and low network bandwidth, making resource-intensive rule-/signature-based systems impractical.
- *Heterogeneous traffic patterns:* Short, periodic flows and diverse communication behaviors place additional strain on real-time detection mechanisms.
- *Conventional detection limitations:* Traditional anomaly detection methods often rely on full-feature extraction and high computational power for inference, thus leading to excessive bandwidth consumption and sluggish responsiveness.

Generally, IoT traffic is characterized by short-lived connections and periodic, repetitive patterns (e.g., 100-byte sensor data sent every 10 min). This differs markedly with the longer-duration, high-volume transmissions in enterprise networks. Moreover, traditional security systems are often resource-intensive, particularly during scans or when applying complex rule sets. They can quickly deplete system resources.

For low-power IoT and edge devices, low-rate distributed denial of service (DDoS) attacks targeting IoT networks[5] represent a growing threat that requires further investigation.[14] For a resource-constrained environment, we propose a lightweight and efficient real-time anomaly detection solution, including identifying bandwidth-draining anomalies.

Deep learning shows promise in traffic anomaly detection. However, models like long short-term memory (LSTM), Slim Transformer, and recent state-of-the-art (SOTA) architectures [Mamba[16] and Convolutional Kolmogorov–Arnold Networks (CKAN)[15]] struggle on

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constrained IoT edges. They either impose excessive computational burdens via complex state-space/spline evaluations or fail to capture short-burst temporal dependencies. Fortunately, GRUs solve this by efficiently capturing temporal DDoS bursts with fewer parameters and lower latency.

Some existing performance limitations include:

- *Lack of IoT optimizations:* Current extended Berkeley packet filter (eBPF) solutions target enterprise networks, lacking strict memory bounding (e.g., hash map limits) to prevent edge out-of-memory crashes.
- *Heavy model overhead:* Complex models like transformers exceed edge computational limits. Even with optimizations like compression[12] or graph enhancements,[13] they struggle to meet the strict memory and latency constraints of IoT gateways.
- *Inflexible static feature sets:* Hard-coded features derived from single datasets fail to generalize across heterogeneous IoT protocols. Furthermore, high-dimensional static sets (e.g., 77 dimensions) cause severe transmission bottlenecks in narrowband links (e.g., LoRaWAN) and lack adaptability to unseen environments.

Targeting IoT applications, we design a resource-efficient real-time anomaly detection system through integrating an eBPF/XDP (express data path) and GRU model. Via an adaptive two-stage feature selection strategy, high-dimensional feature spaces are dynamically compressed into a minimal optimal subset (typically $\leq$11 dimensions) tailored to the specific network context. The eBPF/XDP processes packets at driver layer, exports to user space via a hash map, and a pretrained GRU performs classification, enabling lightweight traffic anomaly detection. Thus, our key contributions are:

- *Adaptive triple-dataset validation for heterogeneous IoT:* Unlike prior works confined to single environments, we validate our framework across CICDDoS2019, IoT-23, and TON_IoT datasets. Our adaptive algorithm successfully navigates through blind spots (e.g., five-tuple obfuscation in DDoS) to achieve a highly stable SOTA accuracy of ≈93% across all three distinct IoT landscapes, demonstrating exceptional robustness and generalizability.
- *A generic adaptive feature selection engine:* We propose and rigorously validate a novel, adaptive two-stage feature screening strategy (Gini Gain prescreening combined with GRU-based incremental evaluation). This generic engine serves as the core of our framework, automatically identifying the minimal yet sufficient feature subset (e.g., compressing raw dimensions by up to 93.5% to five to nine core features) for any given IoT environment, as evidenced by our ablation studies across three distinct benchmarks.
- *Ultralightweight end-to-end edge architecture:* We designed a unified "eBPF/XDP collection + user-space GRU inference" architecture strictly bounded by IoT resource limits. Quantitative software profiling reveals that the system operates with microsecond-level kernel interception (1.85 μs) and millisecond-level inference (10.38 ms), while consuming only 0.67% CPU and a fixed 5.63 MB memory footprint on a Raspberry Pi 4B. Compared to heavy transformer baselines, our GRU model achieves superior edge-deployment feasibility for standard IoT gateways.

Our research roadmap and organization is as follows: To systematically address the aforementioned IoT challenges, we structured our research methodology as a progressive pipeline. First, we conduct a comparative study to identify GRU as the most suitable lightweight sequence model for edge deployment (Step 1: Model selection, detailed in the "Hardware Deployment and System Performance" section). We then leverage eBPF/XDP for kernel-level, ultraefficient traffic feature extraction (Step 2: Data acquisition, reviewed in the "Background" section). The core of our work is the adaptive feature selection framework that bridges the data and model, automatically tailoring the feature set to any specific IoT environment (Step 3: Adaptive core, detailed in the "Adaptive Feature Selection Framework" section). Finally, we validate the generalizability of this integrated framework across three diverse IoT benchmarks ("Framework Validation in Heterogeneous IoT Environments" section), and demonstrate its real-time feasibility on resource-constrained hardware ("Hardware Deployment and System Performance" section).

BACKGROUND

To contextualize the proposed eBPF/XDP-integrated deep learning framework for IoT, this section reviews core technologies (eBPF, XDP, GRU), and IoT network characteristics, all informing the design for resource- and bandwidth-constrained IoT environments targeting lightweight real-time anomaly traffic detection, which is a critical need highlighted in recent surveys on IoT security.[14]

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IoT Network Characteristics and Challenges

Unlike enterprise environments, IoT networks impose unique constraints on anomaly detection systems:

- *Resource limits:* Edge devices (e.g., ARM Cortex-A53) feature limited CPU (<1 GHz) and RAM (<512 MB), alongside strict battery constraints. This precludes heavy models (e.g., ResNet1D) and necessitates low-overhead, memory-bounded processing (e.g., eBPF hash maps, XDP, GRUs).
- *Bandwidth scarcity:* Narrowband protocols (e.g., LoRa/NB-IoT, 1–250 kbps) make transmitting raw traffic or 77-dimensional features nearly infeasible (e.g., transmitting 1 MB of features takes 8000 s on a 1-kbps link).
- *Traffic specificity:* IoT traffic involves short, periodic connections (e.g., 100 bytes/10 s) and heterogeneous protocols (MQTT, CoAP). Threats typically target availability (DDoS) or data theft, demanding customized, highly efficient detection logic.

GRU for Lightweight Sequence Modeling

The GRU efficiently captures critical temporal dependencies in network traffic.[9] By merging LSTM's forget and input gates, GRUs minimize parameters and memory, enabling robust, millisecond-level inference on constrained IoT edge gateways. While advanced models like Mamba[16,17] and CKAN[15] excel in long-sequence modeling and nonlinear mappings, their intricate computations overcomplicate edge deployment. Ultimately, GRU achieves a Pareto-optimal balance, delivering essential temporal memory with a minimalist hardware footprint.

eBPF and XDP

Evolving from classic BPF,[3,4] eBPF introduces a sandboxed kernel virtual machine executing programs on kernel events.[6] Built upon eBPF, XDP attaches directly to the network driver's reception path.[1] Together, they intercept packets before the protocol stack, enabling in-kernel processing without expensive user-space memory copies.

For IoT environments, this architecture provides two critical advantages:

- *High throughput:* XDP processes up to 26 million packets/second/core,1 accommodating bursty telemetry data.
- *Resource efficiency:* Bypassing the kernel stack minimizes CPU overhead for battery-powered sensors.

By filtering irrelevant traffic and extracting metadata directly at the link layer, the eBPF/XDP synergy forms the ideal low-overhead foundation for real-time IoT anomaly detection.

These characteristics mandate lightweight, bandwidth-efficient, IoT traffic-specific detection systems, core design tenets of the proposed eBPF/XDP-GRU framework, meeting the lightweight real-time IoT anomaly detection objective.

ADAPTIVE FEATURE SELECTION FRAMEWORK DESIGN

To address the extreme heterogeneity of IoT protocols and the scarcity of edge resources, relying on static, hard-coded feature sets is insufficient. We propose a generic adaptive feature selection framework that functions as a middleware between raw traffic ingestion and threat detection. This framework is designed to automatically identify the optimal, minimal feature subset $\mathcal{S}_{\text{opt}}$ from an arbitrary high-dimensional feature space $\mathcal{F}$ for any given IoT environment $\mathcal{D}$.

Framework Overview

The framework operates on a "context-aware" principle, functioning as a middleware between raw traffic ingestion and threat detection.

Instead of relying on a static architecture, we design a dynamic, closed-loop workflow that bridges the gap between offline learning and online enforcement. As illustrated in Figure 1, the system integrates the optimized architectural components—specifically eBPF/XDP for the Data Plane and GRU for the control plane—into a unified lifecycle containing two phases:

- *Phase A: Offline adaptive learning:* Illustrated in the left section of Figure 1, this phase functions as the system's "brain." It utilizes information theory to eliminate statistically irrelevant noise from the heterogeneous raw data and employs the Context-Aware Algorithm (Algorithm 1) to output the optimal feature subset ($\mathcal{S}_{\text{opt}}$) and the pretrained model ($\mathcal{M}$).
- *Phase B: Online real-time detection:* Illustrated in the right section of Figure 1, this phase represents the runtime architecture. The red dashed lines indicate the configuration flow: the eBPF programs in the kernel space are dynamically configured to extract only the features defined in $\mathcal{S}_{\text{opt}}$, while the user space inference engine loads the weights from $\mathcal{M}$ to execute real-time detection (Algorithm 2).

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Let $\mathcal{D}$ be an arbitrary IoT traffic dataset containing N flows, and $\mathcal{F} = \{f_1, f_2, \ldots, f_M\}$ be the raw feature set (e.g., $M = 77$ for CICDDoS2019, $M = 18$ for IoT-23, or $M = 44$ for TON_IoT, as detailed in the "Framework Validation in Heterogeneous IoT Environments" section). The objective is to find a subset $\mathcal{S}_{\text{opt}} \subset \mathcal{F}$ such that $|\mathcal{S}_{\text{opt}}| \ll M$ while maximizing the detection accuracy.

Phase I: Information-Theoretic Prescreening

In high-dimensional IoT traffic data, many features are either redundant or have zero variance (e.g., constant IP headers). We employ Gini Importance to quantify the global discriminative power of each feature. For a feature $f_i \in \mathcal{F}$ and a dataset $\mathcal{D}$ with C classes, the Gini Impurity is defined as

$$\text{Gini}(\mathcal{D}) = 1 - \sum_{k=1}^{C} p_k^2 \quad (1)$$

The Gini Gain for a feature f_i is calculated as the reduction in impurity

$$\Delta\text{Gini}(f_i) = \text{Gini}(\mathcal{D}) - \sum_{v \in \text{values}(f_i)} \frac{|\mathcal{D}_v|}{|\mathcal{D}|} \text{Gini}(\mathcal{D}_v). \quad (2)$$

Features are ranked by $\Delta\text{Gini}(f_i)$ in descending order. We retain a candidate set C consisting of the top-K features (e.g., $K = 20$) to filter out noise while preserving potential behavioral signals.

Phase II: Context-Anchored Incremental Augmentation

The core innovation of our framework lies in this phase. Unlike traditional wrapper methods that search blindly, we introduce a context-anchored strategy.

For *initialization with context*, the algorithm first checks the availability of explicit device fingerprints (context features), specifically source/destination ports:

- *If Ports are reliable* (e.g., IoT Local LAN), the base set $\mathcal{S}_{\text{base}}$ is initialized with {*Protocol*, *IPs*, *Ports*}.
- *If Ports are obfuscated/unavailable* (e.g., encrypted tunnels or core gateways), $\mathcal{S}_{\text{base}}$ falls back to {*Protocol*}.

For incremental learning, starting from $\mathcal{S}_{\text{base}}$, the framework iteratively tests features from the candidate

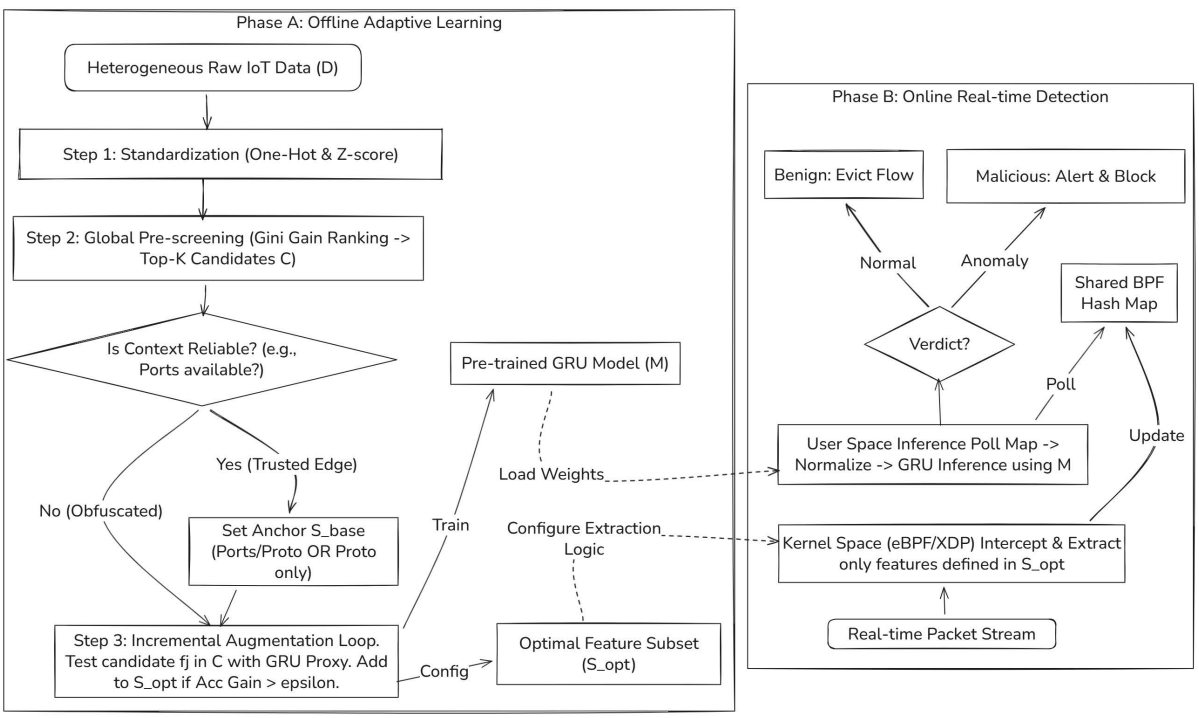


FIGURE 1. End-to-end workflow of the proposed adaptive framework. *Phase A* illustrates the offline process of adaptively selecting optimal features ($\mathcal{S}_{\text{opt}}$) and training the GRU model based on environment context (Algorithm 1). Phase B shows how the selected features and trained model are deployed for synergistic kernel-user space real-time detection (Algorithm 2). The dashed lines indicate the configuration flow from offline learning to runtime deployment.

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set C using a lightweight GRU proxy model. A feature $f_j \in C$ is permanently added to $\mathcal{S}_{\text{opt}}$ if and only if

$$\text{Acc}(\mathcal{S}_{\text{curr}} \cup \{f_j\}) - \text{Acc}(\mathcal{S}_{\text{curr}}) > \epsilon \quad (3)$$

where ϵ is a strict improvement threshold (set to 0.001). This ensures that the framework only incorporates statistical features (e.g., packet length variance) when they function as necessary compensations for missing context.

The complete generic procedure is detailed in Algorithm 1.

Once the optimal subset $\mathcal{S}_{\text{opt}}$ is identified via the training phase, it is deployed into the real-time detection pipeline. The synergistic execution flow between the kernel-space data plane (for feature extraction) and the user-space inference engine is formally described in Algorithm 2.

Framework Advantage Analysis

This adaptive design offers three critical advantages over static SOTA approaches.

Handling Heterogeneity via Context Awareness

Static feature sets assume a fixed protocol stack. However, IoT environments range from IP-based smart homes to non-IP industrial buses. By decoupling the *base context* from *statistical augmentation*, our framework allows the model to "evolve" distinct detection logic: relying on device fingerprints (ports) at the edge, while shifting to behavioral statistics (e.g., flow bytes/s) in the core network.

Feature Compensation Mechanism

A unique capability of our framework is *feature compensation*. In scenarios where key metadata is obfuscated (e.g., five-tuple spoofing in DDoS), the incremental evaluation phase automatically "discovers" high-dimensional statistical proxies (e.g., *Packet Length Min*) to compensate for the information loss. This robustness is mathematically guaranteed by the greedy maximization logic in Algorithm 1.

Minimalist Resource Footprint

By enforcing a strict improvement threshold ϵ, the algorithm aggressively prunes redundant features. This results in a sparse feature vector (typically <10 dimensions), which is crucial for:

- *Bandwidth:* Reducing telemetry overhead on narrowband IoT links (LoRa/NB-IoT).
- *Memory:* Preventing eBPF map overflows on constrained gateways (e.g., Raspberry Pi).

The Core: A Generic Feature Selection Engine

It is crucial to emphasize that the primary contribution of this work is not a static, "one-size-fits-all" feature set. Instead, we propose a generic adaptive feature

Algorithm 1. Generic Adaptive Feature Selection Framework

Input: Arbitrary IoT Dataset $\mathcal{D}$, Raw Feature Set $\mathcal{F}$
Output: Optimal Feature Subset $\mathcal{S}_{\text{opt}}$

Phase 1: Standardization and Pre-screening
Standardize $\mathcal{D}$ using *Z*-score Normalization to obtain $\mathcal{D}_{\text{norm}}$
for each feature $f_i \in \mathcal{F}$ **do**
  Calculate Gini Gain $G(f_i)$ using Decision Tree on $\mathcal{D}_{\text{norm}}$
end for
Rank $\mathcal{F}$ by $G(f_i)$ in descending order
Select Candidate Set $C \leftarrow \text{Top-}K(\mathcal{F}) \setminus \mathcal{S}_{\text{base}}$
Phase 2: GRU-based Incremental Evaluation
Context-Anchored Initialization: Dynamically construct the base feature set $\mathcal{S}_{\text{base}}$ based on the metadata availability in $\mathcal{D}$:
- If *Source/Destination Ports* are reliable and discriminative (e.g., in IoT-23 or TON_IoT), $\mathcal{S}_{\text{base}} \leftarrow \{Protocol, Ports\}$.
- If *Ports* are obfuscated or unavailable (e.g., in CICDDoS2019), $\mathcal{S}_{\text{base}} \leftarrow \{Protocol\}$.

(Note: IP addresses are intentionally excluded from $\mathcal{S}_{\text{base}}$ due to high cardinality and privacy concerns.)

Initialize $\mathcal{S}_{\text{opt}} \leftarrow \mathcal{S}_{\text{base}}$, $Acc_{\text{max}} \leftarrow \text{Evaluate}(\mathcal{S}_{\text{base}})$
while $C \neq \emptyset$ **do**
  $f_{\text{best}} \leftarrow$ NULL, $Acc_{\text{best_step}} \leftarrow 0$
  for each feature $f_j \in C$ **do**
    $\mathcal{S}_{\text{curr}} \leftarrow \mathcal{S}_{\text{opt}} \cup \{f_j\}$
    Train GRU model using $\mathcal{S}_{\text{curr}}$ and get validation accuracy Acc_{val}
    if $Acc_{\text{val}} > Acc_{\text{best_step}}$ **then**
      $Acc_{\text{best_step}} \leftarrow Acc_{\text{val}}$, $f_{\text{best}} \leftarrow f_j$
    end if
  end for
  if $(Acc_{\text{best_step}} - Acc_{\text{max}}) > 0.001$ **then**
    $\mathcal{S}_{\text{opt}} \leftarrow \mathcal{S}_{\text{opt}} \cup \{f_{\text{best}}\}$
    $Acc_{\text{max}} \leftarrow Acc_{\text{best_step}}$
    $C \leftarrow C \setminus \{f_{\text{best}}\}$
  else
    break
  end if
end while
return $\mathcal{S}_{\text{opt}}$

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Algorithm 2. Real-Time Detection based on Selected Features

Input: Packet Stream $\mathcal{P}$, Optimal Features $\mathcal{S}_{opt}$, Pre-trained Model $\mathcal{M}$
Output: Real-time Verdict and Mitigation

Stage 1: Kernel-Space Processing (eBPF/XDP)
for each packet $pkt \in \mathcal{P}$ **do**
  Extract Header Context $\mathcal{K}$ (e.g., five-tuple)
  Update Flow Statistics in Shared Map (e.g., bytes, packets)
  if Feature $f \in \mathcal{S}_{opt}$ is calculable in Kernel **then**
    Compute f and update Map entry
  end if
  return XDP_PASS {zero-copy analysis}
end for
Stage 2: User-Space Inference Loop
while System Running **do**
  for each flow F in Shared Map **do**
    Retrieve feature vector $\mathbf{x}$ corresponding to $\mathcal{S}_{opt}$
    $\mathbf{x}_{norm} \leftarrow (\mathbf{x} - \mu)/\sigma$ {Z-score Normalization}
    Probability $\leftarrow$ Inference $(\mathcal{M}, \mathbf{x}_{norm})$
    if ArgMax (*Probability*) == BENIGN **then**
      Evict flow F from Map {Benign-First Eviction}
    else
      Trigger Alert and Retain for Forensics
    end if
  end for
  Sleep(*Interval*)
end while

selection engine. The algorithm (Algorithm 1) is designed as a middleware that can intake any high-dimensional IoT traffic data and automatically distill it into a minimal, context-aware feature signature. This engine's capability to generate divergent, high-performance feature subsets for different environments (as evidenced in the "Framework Validation in Heterogeneous IoT Environments" section) is the definitive proof of its generalizability, moving beyond the limitations of static feature engineering.

FRAMEWORK VALIDATION IN HETEROGENEOUS IoT ENVIRONMENTS

To empirically validate the adaptivity and robustness of the proposed framework, we conducted extensive experiments across three benchmark datasets representing distinct IoT edge scenarios. The goal is to demonstrate that our generic algorithm (see the "Adaptive Feature Selection Framework Design" section) can automatically converge to the optimal feature subset tailored for each specific environment.

Validation Methodology and Datasets

We selected three diverse datasets to simulate the extreme heterogeneity of the IoT landscape:

- *CICDDoS2019 (Enterprise Edge):* Represents high-speed gateway traffic with obfuscated metadata (simulating encrypted or spoofed headers).
- *IoT-23 (Smart Home):* Represents consumer IoT devices (e.g., Hue lamps, Amazon Echo) infected by Mirai/Torii botnets.
- *TON_IoT (Industrial IoT):* Represents Industrial IoT telemetry data (Modbus/MQTT) with data injection attacks.

For standardized preprocessing, consistent with the generic framework, all datasets underwent a unified pipeline: 1) cleaning and aggregation, 2) one-hot encoding for categorical fields (e.g., Conn State, Service) to expand them into sparse binary vectors, and 3) *Z*-score normalization for continuous fields.

Case Study I: Enterprise Edge (CICDDoS2019)

In this scenario, the environment is characterized by volumetric bursts, and we simulated a "context-constrained" setting where port information is unreliable (due to spoofing).

Feature Selection Trajectory

- *Phase I (context anchor):* The framework initialized with a minimal base set: $\mathcal{S}_{base} = \{Protocol\}$. The baseline accuracy was merely 11.94%, confirming that protocol alone is insufficient for anomaly detection.
- *Phase II (adaptive augmentation):* The algorithm automatically compensated for the lack of ports by selecting statistical behavioral features. It adaptively prioritized Packet Length Min (+75.67% gain), followed by Init Fwd Win Bytes (+4.54%). The late inclusion of Avg Fwd Seg Size and flow bytes/s provided final fine-tuning.

Result

The final subset (five features) achieved an accuracy of 92.73%. The selection of Init Fwd Win Bytes is physically significant as it fingerprints the fixed window sizes often hard-coded in automated DDoS attack scripts.

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Algorithm Intelligence

This case perfectly demonstrates the framework's context-aware compensation capability. Faced with obfuscated ports (a context-poor environment), the algorithm correctly ignored them and autonomously constructed a high-performance detection logic solely from statistical behavioral features.

Case Study II: Smart Home (IoT-23)

This scenario involves specific device fingerprints and botnet propagation behavior.

Feature Selection Trajectory

- *Phase I (context anchor):* Given the fixed function of smart devices, ports are reliable fingerprints. Thus, $\mathcal{S}_{\text{base}} = \{Src/DstPort, Protocol\}$. The baseline accuracy started strong at 68.97%.
- *Phase II (adaptive augmentation):* The algorithm focused on connection lifecycles. The algorithm first selected orig_ip_bytes (+18.7% gain), pinpointing the small-packet nature of C&C heartbeats. Subsequently, conn_state_RSTOS0 (+4.4%) was added to capture scanning failures.

Result

Accuracy rose to 92.57%. The selection of *conn_state_RSTOS0* (connection reset after SYN) precisely captures the "horizontal scanning" phase of Mirai botnets, distinguishing it from legitimate traffic.

Algorithm Intelligence

This case showcases the framework's context-anchored augmentation mode. Given reliable ports as device fingerprints, the algorithm leveraged them as a strong anchor and then intelligently supplemented them with minimal behavioral features specific to the threat landscape.

Case Study III: Industrial IoT (TON_IoT)

This scenario features strictly periodic telemetry data.

Feature Selection Trajectory

- *Phase I (context anchor):* Similar to IoT-23, ports were used as anchors. Baseline accuracy was 62.33%.
- *Phase II (adaptive augmentation):* The selection trajectory was dominated by volumetric features: dst_ip_bytes (+18.3%), src_ip_bytes (+9.6%), and packet counts (src_pkts). This multidimensional focus on payload volume aligns perfectly with detecting Data Injection attacks in periodic industrial telemetry.

Result

Accuracy peaked at 93.49%. The focus on byte/packet counts aligns with the physical nature of data injection attacks, which disrupt the strict periodicity and payload size of industrial sensors.

Cross-Case Comparative Analysis

The power of the framework is best illustrated by comparing the automatically selected feature subsets side-by-side (Table 1).

The "Grand Slam" of Adaptivity: Divergence over Convergence

Table 1 provides definitive evidence of the framework's core intelligence: It does not converge to a universal feature set, but rather diverges to create distinct, environment-specific detection signatures.

- *Compensation mode (CICDDoS2019):* Faced with obfuscated ports (context-poor), the framework correctly disregards them and constructs a new detection logic purely from statistical behavior, like *Packet Length Min* and *Init Fwd Win Bytes*.
- *Fingerprinting mode (IoT-23/TON_IoT):* Given reliable ports (context-rich), the framework leverages them as anchors and then augments them with complementary behavioral features specific to the threat (e.g., *conn_state_RSTOS0* for botnet scanning).

Empirical Proof Against "Universal Features"

The radical difference between the final optimal subsets $\mathcal{S}_{\text{opt}}$ for each case study—ranging from port-centric to purely statistical—constitutes direct, empirical proof that a static, one-size-fits-all feature set cannot exist in the heterogeneous IoT landscape. The framework's ability to autonomously navigate this divergence and craft a high-performing, minimalist signature for each environment is its defining adaptive capability.

Overall Compression and Performance Summary

The adaptive framework demonstrates consistent and significant dimensionality reduction across three heterogeneous IoT environments without compromising detection accuracy:

- *CICDDoS2019 (Edge Gateway):* 77→5 features (93.5% reduction) with 92.73% accuracy
- IoT-23 (Smart Home): 18→6 features (66.7% reduction) with 92.57% accuracy
- *TON_IoT (Industrial IoT):* 44→9 features (79.5% reduction) with 93.49% accuracy.

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TABLE 1. Step-by-step feature selection trajectory: From context-anchored baseline to optimal subset, demonstrating the self-evolving capability of our adaptive framework across heterogeneous environments.

Dataset (Scenario)	Phase	Selected Feature (One-Hot Encoded)	Physical Interpretation (Why Selected?)	Accuracy	Gain	Reduction
CICDDoS2019 (Edge Gateway) raw dimensions: 77	Phase I	`Protocol`	Base context (ports obfuscated	11.94%	—	**93.5%** five features selected
	Phase II-1	`Packet Length Min`	Captures small-packet flooding (Volumetric)	87.61%	**+75.67%**	
	Phase II-2	`Init Fwd Win Bytes`	Fingerprints automated attack scripts	92.16%	+4.54%	
	Phase II-3	`Avg Fwd Seg Size`	Indicators of TCP segmentation anomalies	92.54%	+0.39%	
	Final	+ *Flow Bytes/s*	Overall traffic throughput anomaly	**92.73%**	+0.19%	
IoT-23 (Smart Home) raw dimension: 18	Phase I	`Src/Dst Port , Protocol`	Device fingerprinting (ports available)	68.97%	—	**66.7.5%** six features selected
	Phase II-1	`orig_ip_bytes`	Captures C&C heartbeat (small periodic payloads)	87.68%	**+18.71%**	
	Phase II-2	`conn_state_RSTOS0`	Identifies scanning failures (Mirai behavior)	92.12%	+4.44%	
	Final	+ *resp_ip_bytes*	Response payload anomalies	**92.57%**	+0.45%	
TON_IoT (Industrial IoT) raw dimension: 44	Phase I	`Src/Dst Port , Protocol`	Service anchor (Modbus/MQTT ports)	62.33%	—	**79.5%** nine features selected
	Phase II-1	`dst_ip_bytes`	Detects data injection (payload size anomaly)	80.69%	**+18.36%**	
	Phase II-2	`src_ip_bytes`	Source-side injection detection	90.26%	+9.57%	
	Phase II-3	`conn_state_OTH`	Abnormal protocol interaction states	91.9%	+1.64%	
	Phase II-4	`src_bytes`	Global source volume anomaly	92.5%	+0.6%	
	Phase II-5	`src_pkts`	Frequency violations (Polling interval)	92.95%	+0.45%	
	Final	+ *dst_bytes*	Global destination volume anomaly	**93.49%**	+0.54%	

For the numbers in bold: **Accuracy**—The final classification accuracy achieved using the optimal selected feature subset for each respective dataset scenario. **Gain**—The maximum accuracy increment (improvement) contributed by a single feature during the adaptive selection process for each dataset. **Reduction**—The overall dimensionality reduction rate achieved by our framework for each dataset.

Collectively, the framework achieves an average accuracy of 92.93% using only five to nine adaptively selected features, proving that aggressive, context-aware feature compression is both feasible and effective for IoT anomaly detection.

Conclusion of Validation

These results collectively certify that our framework is not a static detector but a *generic feature selection engine*, capable of automatically tailoring its detection strategy to the context of any given IoT environment.

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TABLE 2. Comparison of deep learning models on test metrics and time costs.

Model	Accuracy	Precision	Recall	F1-Score	Train Time (s)	Test Time (s)
GRU	0.933	0.9217	0.9330	0.919	**151.27**	**0.26**
RNN	0.9116	0.9367	0.9116	0.8896	329.64	0.58
MLP	0.9312	0.9374	0.9312	0.9203	238.03	0.45
LSTM	0.9307	0.9392	0.9307	0.9154	246.87	0.40
CNN-LSTM	0.9333	0.9268	0.9333	0.9193	285.05	0.44
Slim Transformer	0.6487	0.8564	0.6487	0.6266	518.75	0.71
ResNet1D	**0.9347**	0.9331	**0.9347**	0.9203	431.39	0.65
CKAN	0.9345	0.9342	0.9345	**0.9224**	754.88	1.21
Mamba-ECANet	0.9333	**0.9453**	0.9333	0.9218	615.71	1.05

The GRU model achieves a balance of high accuracy and stability, while Mamba-ECANet attains the highest precision (0.9453) and CKAN achieves the best overall F1-Score (0.9224). Numbers in bold are the best in performance.

The radical divergence in the final optimal subsets $\mathcal{S}_{\text{opt}}$ across scenarios is not a flaw, but the core evidence of the engine's adaptive intelligence.

HARDWARE DEPLOYMENT AND SYSTEM PERFORMANCE

Following the research roadmap outlined in the first section, we now progress from validation to implementation. First (Step 1: Model selection), we justify the selection of GRU through a comparative study against SOTA deep learning models. Then, we evaluate the end-to-end latency and resource footprint of the full framework on real IoT hardware, validating the feasibility promised by our ultra-lightweight architecture.

Model Selection Justification: Performance Versus Overhead

To identify the optimal inference engine for the adaptively streamlined pipeline of our framework, we conducted a comprehensive comparison of nine diverse deep learning models: GRU, recurrent neural networks (RNN), multilayer perceptron (MLP), LSTM, convolutional neural network (CNN)-LSTM, Slim Transformer, ResNet1D, alongside recent SOTA architectures CKAN[15] and Mamba-ECANet.[17] All models were trained on the preprocessed CICDDoS2019 dataset. We strategically use the CICDDoS2019 dataset (77 features) as a *stress test* to select the most robust yet lightweight model. A model that remains efficient and accurate under this high-dimensional worst-case scenario is inherently suitable for deployment on lower-dimensional IoT datasets (e.g., IoT-23 with 18 features).

As illustrated in Table 2, experimental results confirm GRU as the optimal IoT choice due to its balanced performance:

- *Accuracy (0.9330):* The GRU achieves a test accuracy of 0.933 when trained on all 77 features of the CICDDoS2019 dataset, establishing a performance upper bound. Our adaptive framework achieves a comparable accuracy of 92.73% using only five selected features (see Table 1).
- *Precision (0.9217):* This high precision ensures a low false alarm rate, which is critical for preventing valid IoT control traffic from being blocked.
- *Recall (0.933):* Matching its accuracy, the strong recall indicates a robust capability to identify virtually all malicious traffic instances.
- *F1-score (0.919):* Reflecting a stable balance between precision and recall, GRU outperforms RNN (0.8896) and the Slim Transformer (0.6266).
- *Computational efficiency:* Crucially, GRU boasts the lowest computational overhead among all evaluated models, requiring only 151.27 s for training and a mere 0.26 s for testing. This ultra-fast inference latency is paramount for real-time eBPF interception at the edge.

This comparative analysis justifies the selection of GRU as the optimal inference engine for our subsequent lightweight, edge-deployment-focused framework. In contrast, other models fail to meet the specific constraints of IoT environments:

- *MLP and CKAN:* (accuracy ≈0.93) show competitive static scores. However, like MLP, the SOTA nonlinear approximator CKAN[15] treats flow features as isolated points, lacking the recurrent structure needed to capture temporal dependencies (e.g., distinguishing periodic sensor data from low-rate DDoS). Furthermore, CKAN incurs the highest computational burden (754.88 s

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TABLE 3. System performance profiling under DDoS Stress on Raspberry Pi 4B (using CIC-DDoS2019 Dataset)

Category	Performance Metric	Measured (Avg)	Peak Value	Observation
Latency	Kernel Interception (T_{kernal})	1.85 μs	20.11 μs	Stable microsecond plane
	GRU Inference (T_{user})	10.38 ms	57.89 ms	Real-time classification
Resources	Agent CPU utilization	0.67%	2%	Near-zero compute cost
	Memory RSS footprint	5.63 MB	5.63 MB	Deterministic stability

training, 1.21 s test time), rendering it impractical for edge deployment.

- *Mamba-ECANet, CNN-LSTM, and ResNet1D* provide structural or temporal inductive biases, but sequence-based SOTA models, like Mamba-ECANet[17] and deep convolutional networks incur higher parameter scales and prolonged inference times. Specifically, Mamba-ECANet's test time (1.05 s) is exactly four times that of GRU, a latency penalty that exceeds the strict resource limits of IoT gateways.
- *Slim Transformer* performs poorly (accuracy 0.6487, F1 0.6266). We attribute this to a fundamental mismatch between the transformer's inductive bias and the characteristics of our IoT traffic data. Transformers excel in domains with large-scale pretraining data and long-range contextual dependencies. However, our IoT traffic datasets (e.g., CICDDoS2019) represent small-sample, high-dimensional, and sparse sequential data typical of edge environments. Without massive pretraining, the transformer's complex multihead attention mechanisms are prone to overfitting on these limited samples, failing to generalize. In contrast, GRU's recurrent inductive bias (gate mechanisms for short-term memory) is inherently more suited for modeling the short-burst temporal patterns of network flows. While recent work on graph-enhanced transformers for multivariate IoT data[13] shows promise, such models still face prohibitive computational overhead for real-time inference on resource-constrained edge gateways, like Raspberry Pi 4B. Our lightweight GRU, paired with adaptive feature selection, provides a far more practical and effective tradeoff for edge deployment.

End-to-End Hardware Performance Analysis

To rigorously validate real-world performance, we deployed the full framework on a Raspberry Pi 4B (Quad-core Cortex-A72 @ 1.5 GHz, 4GB RAM) serving as an IoT Gateway. We replayed attack traffic from the CIC-DDoS2019 dataset under sustained stress for 100 s and monitored system resources in real-time. The results, summarized in Table 3, confirm the framework's ultralightweight nature and deterministic resource behavior. We selected the CIC-DDoS2019 dataset for hardware profiling due to its volumetric complexity and representativeness of high-throughput edge gateway traffic, providing a stringent testbed to demonstrate the framework's capability under stress.

Microsecond-Level Kernel Processing (T_{kernal})

The eBPF/XDP program attached to the driver hook processes each packet with an average latency of only 1.85 μs. Occasional peaks up to 20.11 μs correspond to the atomic initialization of new flow entries in the BPF hash map. This performance demonstrates wire-speed processing capability, ensuring security inspection does not become a bottleneck.

Millisecond-Level User-Space Inference (T_{user})

The GRU model, operating on the adaptively selected minimal feature set (five features for CICDDoS2019), performs inference in 10.38 ms on average per flow, with occasional peaks up to 57.89 ms during system initialization or high-load conditions. This efficiency makes real-time classification feasible on edge hardware.

Negligible Resource Footprint

Our monitoring shows that the entire defense agent consumes a mere 0.67% of the CPU on average during attack replay. Crucially, the memory footprint remains an absolutely deterministic 5.63 MB (resident set size), unaffected by traffic volume. This stability is guaranteed by our eBPF map design and benign-first eviction policy, preventing memory exhaustion, a critical requirement for constrained devices.

In summary, the profiling data conclusively prove that our framework meets the stringent demands of IoT edge deployment. It delivers "microsecond-level" packet processing (1.85-μs average, 20.11-μs peak), "millisecond-level" AI inference (10.38-ms average, 57.89-ms peak), and a "near-zero, deterministic resource footprint" (0.67% CPU, 5.63 MB memory), fulfilling the

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promise of a truly lightweight and real-time anomaly detection system.

CONCLUSION AND FUTURE WORK

In this article we propose a highly scalable, adaptive real-time anomaly detection framework integrating eBPF/XDP driver-based traffic processing with lightweight GRU inference. By replacing static, hard-coded feature sets with an intelligent adaptive feature selection mechanism, the system dynamically identifies the most discriminative behavioral characteristics for heterogeneous IoT environments.

Extensive evaluations across three diverse benchmarks (CICDDoS2019, IoT-23, and TON_IoT) demonstrate that the framework successfully handles the high-dimensional sparse feature space expanded by one-hot encoding, consistently compressing inputs by up to 93.5% to just five to nine core behavioral features while maintaining a stable SOTA accuracy of ≈93% across all scenarios. Hardware profiling on a Raspberry Pi 4B confirms microsecond-level kernel processing (1.85 μs), millisecond-level inference (10.38 ms), and a negligible resource cost (0.67% CPU, 5.63 MB memory), completely neutralizing performance bottlenecks in resource-constrained edge gateways.

Future work will focus on three directions: 1) full-scale deployment across heterogeneous edge clusters (e.g., Raspberry Pi 4B, Jetson Nano), 2) expanding the adaptive engine to support emerging 5G/6G protocols (e.g., QUIC, SRv6), and 3) integrating lightweight, graph-based multisensor correlation analysis[13] while preserving the framework's ultralow resource footprint.

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